5

The triumph of the aggregate

Knowledge claims and attributions

Modern psychology may have started out with at least three different types of investigative practice, but, as we have seen, one of these hardly survived the first few decades in the history of the discipline. Among the other two, it was the Galtonian type that was to achieve a dominant position during the first half of the twentieth century. In the following chapters we will have to analyze various aspects of this complex historical development. Our analysis will draw heavily on the information contained in the psychological journals that now provide the major vehicle for the establishment of psychological knowledge claims. Twentieth-century psychology is a system of "public knowledge,"[1] and the major disciplinary journals constitute the crucial link in the transformation of the local knowledge produced in research settings into truly public knowledge.

This transformation involves the preparation of a research report that meets the criteria and conventions that prevail within the discipline or research area. Ostensibly, the research article communicates certain "findings" held to represent knowledge by the conventions of a particular community. In fact, the contents of the article only represent a knowledge *claim,* which may or may not be accepted by its intended audience. To convince this audience, the authors must submit to its conventions. For instance, they must adopt a certain literary style, relate the contents of the article to known concerns and interests of its possible audience, demonstrate the conformity of their methods to prevailing norms, and so on.[2] But the centerpiece of its work of persuasion is likely to involve the "findings" themselves, for they are after all the raison d'être of the whole paper. If the "results" that the author(s) wish to communicate do not measure up to prevailing standards of what constitutes significant knowledge in their chosen field, the report will not see the light of day, no matter how immaculate its style might otherwise be. These standards, however, are not eternal verities but are historically variable. Individual researchers must take these standards as they find them, but the potential contribution of a historical perspective lies in uncovering their contingent nature.

Whereas sociological field work has been able to study the way in which individual researchers accommodate themselves to prevailing standards of knowledge production, historical analysis examines changes in these standards themselves. This can be done by following changes in the patterns of reporting practices in selected journals over a period of time.

To establish their knowledge claim, researchers must transform whatever records they have made during the process of investigation into a standard format that is acceptable in their field of work. That is to say, they must "decontextualize" their records of what they did and saw in the laboratory and "recontextualize" this material in accordance with the standards of the audience that will adjudicate their knowledge claim.[3] In the case of psychological research certain aspects of the participants' interaction must be transformed into a symbolic form, which will eventually appear as the "findings" of the investigation. It is this form we now need to examine.

When one examines knowledge claims, as reflected in scientific psychological papers, one finds that they always have a basic propositional form. Some set of attributes is said to be true of certain subjects under certain conditions. What the "results" section of the paper says is that one or more persons did such and such when in a particular situation. The knowledge objects that are the legitimate focus of interest for scientific psychology involve the attributes of a specially constituted subject, namely, the research subject. Most commonly, it is the attributes, rather than the subject whose attributes they are, that are considered to be problematic for psychological research. But if we are to achieve any insight into fundamental changes in the investigative practices of the discipline, we need to reverse this perspective and to recognize the problematic status of the research subjects themselves. For when the nature of the subjects changes, the nature of their attributes changes too. For instance, a subject who is defined as an individual consciousness has a different set of attributes from a subject who is defined as a biological population. If we want to understand the most general historical changes in research practice, we need to deconstruct the research subjects that an army of investigators have constructed in obedience to various conventions about the proper objects of psychological knowledge. Much of the remainder of this book will be devoted to this analysis.

Individual attribution of experimental data

A systematic survey of experimental psychological journals (*American Journal of Psychology, Psychological Review, Philosophische Studien, Psy-*

chologische Studien, Archiv für die gesamte Psychologie) for the period up to World War I shows that in virtually all published studies the experimental results were clearly attributed to individual experimental subjects. Most often the individuals concerned were identified by a name, letter, or initials, but the point worthy of attention is that the form of reporting experimental data involves the attribution of certain "responses" to specific individuals. This pattern is best illustrated by an example from the psychological literature of this period (Table 5.1). Even where results were averaged across a number of subjects, which was often not the case, the responses of each individual were systematically reported. Typically, it was the pattern of individual responses, and not the average result, that formed the basis of the theoretical discussion.

How was this pattern of reporting experimental results related to psychological knowledge claims? Even where the table of results only identified the human data source by a letter or initial, it was usually possible to verify, from other sections of the experimental report, that the individuals to whom the reported responses were attributed were in fact competent and experienced psychological "observers" whose testimony had much greater value than that of naïve subjects. To some extent the knowledge claim undoubtedly rested on the reliability attributed to expert observers. Attaching the name of a reputable observer was a way of establishing the scientific credibility of the data.

However, the individually identified expert research subject could only function in this manner because of the way in which the object of psychological knowledge was defined. In this research tradition it was unquestioned that psychological knowledge had to pertain to the *content of individual minds*. In other words, such knowledge could always be expressed in propositions of which the individual mind was the subject and various statements about mental contents were the predicate. The arrangement of tables of experimental results simply reflected this schema of psychological knowledge by attributing a particular pattern of mental contents to specified individual minds. This represented a new way of constituting psychology's traditional knowledge object, the individual consciousness. Thanks to the new methods, that apparently private object was transformed into an object of public knowledge.

Individual attributions of experimental results did not mean that the phenomena studied were felt to be idiosyncratic. On the contrary, claims for the general or even universal significance of experimental findings were commonly made, but such claims were generally based on one or more additional considerations. For instance, in such areas as the psychology of sensation results from even a single subject could be claimed to have general significance because of the presumed similarity of the underlying physiology in all normal human individuals. Second, claims for generality were supported by repeating the experiment with a few individuals. Each additional

Table 5.1. *A table of results from the* American Journal of Psychology, *1895*

Division of Exps.	OBSERVER, DR. KIRSCHMANN.						OBSERVER, J. O. QUANTZ.					
	LEFT EYE.			RIGHT EYE.			LEFT EYE.			RIGHT EYE.		
	No. of Single Exp.	Av. Deviation from Norm. Mag	M. V.	No. of Single Exp.	Av. Deviation from Norm. Mag	M. V.	No. of Single Exp.	Av. Deviation from Norm. Mag	M. V.	No. of Single Exps.	Av. Deviation from Norm. Mag	M. V.
		° ' "	' "		° ' "	' "		° ' "	' "		° ' "	' "
Equal Discs	50	+0 2 8.7	1 14.92	50	+0 1 10.2	0 40.73	50	+0 0 46.55	0 34.9	50	+0 0 28.7	0 57.05
Unequal Discs	50	+0 2 53.8	1 41.8	50	—0 0 5.5	1 19.25	50	+0 4 58.8	1 16.65	50	+0 3 41.5	1 22.28
Av. for Single Eyes	100	+0 2 31.25	1 28.36	100	+0 0 32.35	0 59.99	100	+0 2 52.68	0 55.78	100	+0 2 5.1	1 9.66

	No. of Single Trials.	Average Deviation.		M. V.	No. of Single Trials.	Average Deviation.		M. V.
		Absolute Value.	Val. Rel. to Norm. Mag.			Absolute Value.	Val. Rel. to Norm. Mag.	
		' "		' "		' "		' "
Total Average	200	+1 31.8	+0.01355	1 14.18	200	+2 28.89	+0.0221	1 2.72

Note: This table is from J. O. Quantz, "The influence of the color of surfaces on our estimation of their magnitude," *American Journal of Psychology* 7 (1895):33. The table is reproduced here merely as an example of personal attribution to named individual observers; the content is not relevant in the present context.

subject then provided material for replication of the original experiment. Third, where attempts at replication were not altogether successful, relevant personal information or introspective evidence could be used to provide a rational explanation of the failures; for example, relevant previous learning experiences or reported fluctuations of attention could be used to account for discrepant results.

There were essentially two versions of this first model of experimentally established psychological knowledge, a strong and a weak version. The strong version was due to the inspiration of Wundt for whom the individual mind was a synthetic unity constituted by real processes of "psychic causality." Mental contents were the attributes of this system and could be used to study its constitution. In the weaker version of some of Wundt's pupils, like Titchener, the reference to real causal processes was dropped, but the individual mind remained the only conceivable locus for mental contents. In fact mental contents retained their status as that which constituted individual minds, although the nature of this constitution was interpreted positivistically rather than in the terms of Wundt's principle of "psychic causality."[4] Throughout this research tradition Wundt's *actuality* principle continued to be respected in principle, if not always in name. That is to say, knowledge of mental contents was interpreted as knowledge of actually occurring events rather than as logical schemes imposed on the products of such events. Whereas the latter approach would not necessarily have required any reference to individual minds, the study of mental contents in their actual occurrence was hardly conceivable without such individual attribution.

Parenthetically, it should be noted that there continued to be a certain analogy between this tradition of psychological experimentation and the tradition of physiological experimentation out of which it had grown. What marked off the relevant domain of physiological experimentation from biochemical or biophysical studies on the one hand, and from biological population studies on the other, was its restriction to the universe of the individual organism.[5] In its most consistent version, this tradition even entailed a commitment to experimentation involving the whole organism. This commitment was explicit in the work of Pavlov, for example, whose experiments on conditioned reflexes provided a kind of objectivistic analogue to the experimental work of mentalistic psychology. For Pavlov the individual living cerebrum was the unit of study, just as the individual active mind was for the experimental psychologists.[6] Both for Wundt and for Pavlov this methodological feature was not just a practical preference. They both saw the purpose of their experiments in terms of an exploration of a very specific system of causal processes, although for Wundt the system was one of psychic causes, whereas for Pavlov it was one of physiological causes. Pavlovian experimentation illustrates the lack of any reciprocal necessity in the connection between the mentalism of early experimental psychology and the individualism of its knowledge claims. Mentalistic knowledge interests may predispose one to individual attribution in one's knowledge claims, but the converse is decidedly not true.

Also, the individual attribution style of making knowledge claims has nothing to do with whether the features that are individually attributed are quantitative or qualitative in nature. In fact, most of the literature with which we are concerned here involves quantitative data. Moreover, individual attribution does not by any means exclude the use of statistics. Systematic psychological experimentation involved use of statistics from the very beginning. This is illustrated by experiments in psychophysics, which were of course a mainstay of the early psychological laboratory. Here, statistical methods were used as a way of combining observations in order to derive an assumed true value still attributable to an individual observer. The calculus of error was applied to a population of observations from a single subject, not to a population of subjects. Any increase in the number of experimental subjects above one constituted a replication of the experiment. If the interindividual variability was large, this was considered prima facie evidence that the attempted isolation of critical determining factors had failed and that uncontrolled disturbing processes had supervened.[7]

Although this early research tradition of modern psychology was constituted by a set of coherent and internally consistent practices, it was apparently inadequate to the changing tasks that psychologists set themselves in the course of the twentieth century. The importance of this style of research practice for the discipline as a whole declined quite rapidly during the early part of the century. At the end of chapter 4 we documented the decline of the Wundtian research paradigm by using the exchange of experimenter and subject roles as a rough index. Before proceeding further we will illustrate this trend by another index that arises out of the preceding analysis of research reports. As we noted in the course of this analysis, the individual attribution of research data was frequently buttressed by identifying experimental subjects by name. We can therefore use the relative prevalence of this practice as an index to demonstrate the progressive decline of this method of establishing psychological knowledge claims. The relevant data are presented in Table 5.2.

It is apparent that the general historical trend parallels the trend illustrated in Table 4.1. In the course of time, experimenter and subject roles are less and less frequently exchanged and research subjects are less and less frequently identified by name. What factors were involved in this decline? That is obviously a very large question, and it cannot be answered by considering the declining research style in isolation from its competitors. The other side of the coin in the relative decline of one set of practices is obviously represented by the relative ascendancy of another set of practices. What we would like to understand is the replacement of one kind

Table 5.2. *Percentage of empirical studies in which all or some of those participating as subjects are identified by name*

Journal title	1894–1896	1909–1912	1924–1926	1934–1936
American Journal of Psychology	54	39	39	24
Psychological Review	35	35	N.C.[a]	
Journal of Educational Psychology	—[b]	4	0	0
Psychological Monographs	—	18	21	5
Journal of Experimental Psychology	—	—	8	8
Journal of Applied Psychology	—	—	4	0

[a]The N.C. entry for the *Psychological Review* indicates the point at which the number of empirical studies published in this journal became too small for meaningful comparison.
[b]Dashes indicate periods before the appearance of a journal.

of investigative practice by another. But this means we have to explore the alternatives more extensively than we have done thus far. One potential basis for an alternative style of psychological research was provided by the pioneering work of Galton, which involved the construction of a different kind of knowledge object whose historical development now needs to be considered more fully.

Alternative basis for knowledge claims: The psychological census

The alternative to the individual attribution of psychological data is to be found in attempts at making psychological knowledge claims by attributing psychological characteristics to collective rather than individual subjects. In other words, one could claim that certain psychologically interesting features were true of some aggregate of individuals. This meant that one's knowledge claims necessarily took a statistical form, the comparison of different aggregates usually being necessary to arouse interest in the data. One could compare different age groups, for example, on some psychological characteristic. Early forms of this sort of statistical comparison were crude, being generally limited to averages, or occasionally, ratios of the population meeting some criterion. The important point was that these research reports would attribute psychological characteristics to groups of subjects rather than to individuals.

Whereas the dominant form of psychological research reporting seemed to take experimental physiology as its model, this alternative style of presenting the results of psychological research appeared to owe more to a practice of quantitative social research that had emerged earlier in the nineteenth century. Indeed, statistical studies of human conduct in the aggregate were both longer and better established than were laboratory studies of psychological processes. Interest in the statistics of crime, suicide, and poverty and in indexes related to public health was quite widespread before the middle of the nineteenth century, both in Europe and in North America. So-called statistical societies, which typically linked the compilation of social statistics to questions of social reform, were active in a number of countries. The numerical depiction of social problems seemed to promise a more "scientific" approach to their solution than the road indicated by political ideology.[8]

On the practical level, the development of social statistics was closely linked to the spreading use of the questionnaire as a method of investigation. Rather than rely exclusively on official statistics the statistical societies began to collect their own information about the living and working conditions of the poor. It was a small step from circulating questionnaires about conditions of child labor to the use of such instruments for the investigation of children in schools.[9] In Berlin, a municipal statistical office was systematically using questionnaires for collecting mass data on children by the 1860s.[10] G. Stanley Hall used the Berlin work as a model in his earliest American investigations of "the content of children's minds."[11] A converging development in the field of medical statistics had led to the compilation of psychological information on children by circulating questionnaires to mothers.[12]

The general popularity of descriptive statistical inquiry in the third quarter of the nineteenth century had made the questionnaire method seem like an appropriate tool for investigating questions of an undoubtedly psychological character. Charles Darwin used it in his study of emotional expression, and his cousin, Francis Galton, used it in his study of heredity and mental imagery.[13] Such usage helped to strengthen the legitimacy of statistical data compiled on the basis of questionnaires as a source of scientific knowledge. This was particularly true of those, like G. Stanley Hall in America, who saw psychology in quasi-Darwinian terms. From this point of view it was important not to restrict psychology to the study of individual minds but to extend it to the distribution of psychological characteristics in populations. Hall gave practical expression to this attitude in the area of child study, of which he was the chief American promoter. In this field the attribution of psychological characteristics to populations, often quite large populations, was common.[14]

But what made the questionnaire acceptable as a tool of psychological research, at least in some quarters, was not just its practical convenience. There was a belief that mass data, gathered by these means, constituted a valid basis for psychological knowledge. This belief was quite new in the latter part of the nineteenth century, and it had needed some profound social and conceptual developments to make it appear plausible. Let us briefly trace some salient aspects of the conceptual background.

The scientific appeal of statistical tables depended on more than the reduction of complex issues to countable "facts." It depended on the numerical regularities that began to appear when the actions of many individuals were aggregated. Rates of crime and suicide, for instance, tended to remain relatively constant for a particular area, and variations among areas could be related to identifiable environmental influences. It was not experimental psychology but the repeated demonstration of striking regularities in social statistics that first convinced a large public that human conduct was subject to quantitative scientific laws.[15] The major methodological implication of these highly effective demonstrations was that the inherent lawfulness of human conduct would become apparent only if observations on a large number of individual cases were combined. This led to an infatuation with large samples, for only through them it seemed could the laws governing human action be made manifest. A felicitous convergence of theoretical and practical interests now became apparent. The collection of large-scale statistical information on human populations had originally been prompted by the administrative concerns of public bureaucracies. Those concerns had dictated an interest in large numbers, and now it began to appear as though large numbers were also necessary to establish the scientific laws of human conduct. Social statistics united the interests of science and administration in a mutually convenient bond, a union that, as we shall see in chapter 7, was to be perpetuated in the psychological statistics of the early twentieth century.

The road that led from social to psychological statistics depended on two fundamental conceptual steps. Quetelet,[16] the pioneer of a statistical social science, took the first step. In order to explain differential but stable rates of social indexes like crime, he invented so-called penchants or propensities which were attributed to an average individual. Thus, differential rates of crime depended on variations in the average "propensity to crime." If rates of crime varied with such factors as age, sex, and climate, this was due to the influence of these factors on the propensity to crime. By analogy, one could think of propensities to suicide, homicide, and insanity. For every such legally or administratively defined category one could imagine a corresponding "propensity." What Quetelet had done was to substitute a continuous magnitude for the distinct acts of separate individuals. Suicide, crime, homicide, and other social acts were not to be understood in terms of the local circumstances of their individual agents but in terms of a statistical magnitude obtained by counting the number of heads and the number of relevant acts in a particular population and dividing the one by the other. The ultimate significance of this step for the methodology of modern psychology cannot be overemphasized.

However, it took a considerable time for this kind of statistical thinking to be directly applied to psychological questions. At first, there was considerable divergence of opinion about what the regularity of social statistics implied for individual conduct. Quetelet attributed the "propensities" to an average individual, not to specific individuals,[17] and German nineteenth-century statisticians argued strenuously against making inferences of individual determination on the basis of regularities in social statistics.[18] In England, on the other hand, a different view rapidly gained ground. Buckle, who was very widely read, was quite insistent that statistical regularities in human populations were clear evidence for the lawfulness of individual actions.[19] Such a claim depended on a very specific view of the relationship between collectivities and their individual members: Not only were individual attributes freely composable into aggregates (implying the continuity assumption), but, conversely, group attributes were to be regarded as nothing but summations of individual attributes. This latter view was consonant with Charles Darwin's contemporaneous conception of a biological species as an interbreeding population of individuals rather than as an essential type.[20]

That such assumptions could provide the basis for a technology of research on individual differences was clearly grasped by Francis Galton. His technical contributions and those of his devoted follower, Karl Pearson, finally made possible a psychology that could plausibly claim to be scientific while not being experimental. A new method for justifying psychological knowledge claims had become feasible. To make interesting and useful statements about individuals it was not necessary to subject them to intensive experimental or clinical exploration. It was only necessary to compare their performance with that of others, to assign them a place in some aggregate of individual performances.[21] Individuals were now characterized not by anything actually observed to be going on in their minds or organisms but by their deviation from the statistical norm established for the population with which they had been aggregated.

As a result of this Galtonian turn, psychology was left with two very different frameworks for justifying its knowledge claims, the one based on an individual attribution of psychological data, the other necessarily relying on statistical norms. Both types of knowledge claim rested on the demonstration of certain regularities that could be interpreted as having psychological significance. But the nature of the regularities was fundamentally different in the two cases. In the case of the traditional paradigm for experimental psychology, as established by Wundt, the regularities demonstrated had an immediate causal significance. One showed how the reactions of an individual psychophysical system changed as the conditions to which it was exposed were varied. The knowledge one obtained referred directly to particular psychophysical systems. But where the established regularities were attributed to groups of individuals, the system to which one's data referred was no particular psychophysical system but a statistical system constituted by the attributes of a constructed collective subject.[22] Whereas the first type of psychological knowledge was similar in principle to the kind of knowledge then obtainable in experimental physiology or even physics, the second type of knowledge seemed to have more in common with census taking and social or biological population studies.

The discrepancy between these alternative kinds of psychological knowledge was enormous. Experimental knowledge claimed its special status on the basis of its similarity to the kind of knowledge on which the older established laboratory sciences had built their success. Its appeal was to its use of skilled observers, unambiguous reactions, and systematically controlled conditions. None of these appeals was available to those who wanted to publish population studies. Quite different criteria for assessing the worth of knowledge claims seemed to operate here. The simple device of multiplying the number of subjects seems to have been resorted to as a compensation for their lack of skill and experience. An intuitive notion of individual deviations canceling out in a large sample probably operated from the start, and in due course this notion was made more precise by the application of the calculus of probabilities.

In the earliest version of the statistical approach to psychological knowledge, the spirit of Quetelet had not been laid to rest. The possibility of deriving *average* values from group data fascinated the early researchers, for these averages were regarded as indexes of some human type, usually an age type or a sex type.[23] The distribution of characteristics in a population was of interest for establishing the existence of common patterns that could be used to define psychological types and subtypes, and also for determining a natural range of observations. Implicitly, the relationship between the individual and the aggregate was still mediated by a type concept, which usually appears in the guise of some supposedly natural, often quasi-biological, category. In this form the statistical approach could hardly compete with the scientific credentials of traditional experimental psychology.

Most psychologists wanted to emulate the harder sciences. Proper scientific knowledge was knowledge pertaining to regular changes in specific psychophysical systems under systematically varied conditions. "Merely" statistical knowledge pertaining to populations was regarded as superficial and incapable of leading to an explanatory science of determinate processes taking place in actual psychophysical organisms.[24] From this point of view the information yielded by population studies was useless for advancing claims to psychological knowledge. The subjects of these studies were children, the most unreliable of observers; their reactions were complex, variable, and unanalyzed, and the conditions under which the information was collected were uncontrolled and open to the influence of unknown factors. Indeed, this kind of study never achieved respectability within the discipline as a whole. Nevertheless, it had some influential supporters within American psychology, including some, like Hall himself and J. McKeen Cattell, who had worked in Wundt's laboratory and were well acquainted with the standard constraints on making psychological knowledge claims. If they gave their support to this alternative style of psychological investigation it was because they were interested in an alternative kind of psychological knowledge. They did not wish to limit such knowledge to an analysis of psychological processes in individual minds but wanted to extend it to include the distribution of psychological characteristics in populations.

For the most part, the early statistical studies established the value of their knowledge claims by an explicit or implicit appeal to a criterion that was unacceptable to traditional experimentalists. This criterion was one of social relevance, of closeness to real life. It operated both for the choice of populations and for the choice of the psychological characteristics attributed to these populations. The groups investigated were defined in terms of culturally or administratively relevant categories, such as boys and girls of a certain age, pupils at certain schools, or Columbia undergraduates. The psychological categories whose distribution was described were also at first taken from everyday life, like the activities of dreaming, playing, reading, writing, and recollecting.[25] Clearly these early population studies sought to legitimate their knowledge claims by appealing to an immediate and obvious practical relevance. They were like a psychological census which, like other kinds of census, might be of use to persons in positions of authority who had to make decisions affecting their charges. Where the appeal of traditional experimentalism was to esoteric scientific norms, the appeal of these statistical studies was to the appearance of practical utility. Except where it appealed through the magical aura of science, the first approach depended on the existence of a rather sophisticated intellectual community, whereas the second approach was more likely to make a direct impact on a lay public.

The grounds for this approach had been prepared by certain historical developments in the second half of the nineteenth century. On the institutional level the educational system had been profoundly transformed into a rationalized system for grading individuals in terms of their ability to measure up to highly standardized social-intellectual demands. This involved new social practices, like standard curricula, age grading, and written examinations, which actually created the kinds of statistical populations that Galtonian psychology took as its basis.[26] The ordering of individuals in terms of their standing in a statistical aggregate was a social task that schools had assumed some time before psychologists were able to show them how to do it better. Of course, for this to be seen as an appropriate task for educational institutions in the first place required specific historical circumstances.

On the ideological level the ground had been prepared by the increasing tendency to conceptualize social problems in terms of populations of individuals. By the closing years of the nineteenth century it was common, especially in the United States, to formulate the human problems of urbanization, industrial concentration, and immigration in terms of the problems of individuals conceived as members of statistical aggregates. Crime, delinquency, feeblemindedness, and so on were easily attributed to the statistical distribution of certain individual characteristics. That meant the transformation of structural social problems into the problems of individuals, which were to be dealt with not by social change but by administrative means. For those psychologists who were prepared to redefine the nature of their subject matter this social context opened up enormous possibilities for establishing a new science that was socially, in the sense of administratively, relevant.[27] For by studying the distribution of psychological characteristics in populations, psychologists might make an important contribution to the management of any social problems that could be plausibly defined in terms of the statistical variation of individual traits.

The young discipline of scientific psychology therefore had to contend with divergent methodological pulls from the very beginning. The norms of laboratory experimentation, as established particularly by Wundt's example, supported psychology's claim to be able to produce knowledge that was comparable with the knowledge produced by the prestigious older experimental sciences. But for this kind of psychological knowledge there was at best a limited demand in the United States.[28] With no apparent practical application, it was difficult to mobilize resources for its support. The kind of knowledge that was in demand had to have clear social relevance, which meant that it had to be applicable to salient social categories and to practical concerns. This kind of knowledge was available, but it could never hope to pass muster before a scientific tribunal. Faced with these opposing demands from an expert and a lay public, the discipline was able, up to a point, to have it both ways. It was sometimes possible to use the appeal of socially relevant psychological knowledge to obtain support for irrelevant laboratory research, and conversely, it was often the case that the magical appeal of laboratory rituals could be used to legitimize false claims to scientific authority by the purveyors of various practical nostrums. But these were essentially short-term expedients that did nothing to improve an inherently unstable situation. In the long term the fortunes of the discipline would depend on its ability to bridge the gap between the two kinds of knowledge claim it was interested in making.

Transition to aggregate data

By 1914, which conveniently marks the end of a period, American psychology contained some rather divergent investigative practices. In particular, there was the opposition between the more traditional attribution of research data to individuals and the more recent practice of attributing it to groups. The question now arises of how these practices were distributed over sections of the discipline and how they fared during the next historical period.

To document these issues, a content analysis of four psychological journals was undertaken covering the period from 1914 to 1936. Articles reporting empirical research on human subjects were categorized in terms of whether the reported data were attributed essentially to individuals or essentially to collective subjects, or whether a mixture of both types of attribution was used.[29]

The five journals from which research reports were sampled included the two most prominent journals of basic experimental research (*American Journal of Psychology* and *Journal of Experimental Psychology*) and the two most important journals devoted to so-called applied research (*Journal of Applied Psychology* and *Journal of Educational Psychology*). As explained in the appendix, all empirical studies that appeared in three consecutive volumes of the journal for each of three time periods were analyzed. These time periods were the mid-1910s, the mid-1920s, and the mid-1930s.[30] Thus, for each of the four journals nine volumes were analyzed in sets of three, spaced roughly a decade apart. The results for each set of three volumes were combined so as to represent a time period rather than a specific year. The percentage of studies using individual, group, or mixed data was then calculated for each time period. The resulting information is presented in Tables 5.3 and 5.4.

Two patterns stand out clearly. The first is a very consistent tendency for group data to appear relatively more frequently in the journals of "applied" research; the second is an overall trend for the use of group data to increase and that of individual data to decrease over the period under review. This latter trend emerges in all five journals, although because the use of group data in the "applied" journals is already very high to begin with, the change is relatively greater in the journals of basic experimental research.

There also appear to be some consistent differences within the group of "applied" journals and within the group of basic journals. The *Journal of Educational Psychology* manifests the overall trend in a slightly more extreme way than the *Journal of Applied Psychology*. In a parallel, though more pronounced way, the *Journal of Experimental Psychology* is always somewhat ahead of the *American Journal of Psychology* in following the prevailing trend. If we use the overall trend as a standard, the *Journal of Educational Psychology* could be characterized as the most progressive of these journals and the *American Journal of Psychology* as the most conservative.

During the historical period between World War I and World War II,

Table 5.3. *Percentage of empirical studies reporting individual or group data: Three journals of basic research*

Type of data	*American Journal of Psychology*	*Psychological Monographs*	*Journal of Experimental Psychology*[a]
1914–1916			
Individual data only	70	62	43
Individual and group data	5	11	19
Group data only	25	27	38
1924–1926			
Individual data only	60	31	41
Individual and group data	6	24	15
Group data only	35	45	44
1934–1936			
Individual data only	31	31	28
Individual and group data	14	16	11
Group data only	55	53	61
1949–1951[b]			
Individual data only	17	3	15
Individual and group data	3	3	2
Group data only	80	94	83

[a]The *Journal of Experimental Psychology* only began to appear in 1916; the analysis for 1914–1916 is based on the first three volumes (actually, 1916–1920).
[b]To avoid the anomalous situation at the end World War II, the last time interval was increased from ten to fifteen years.

the original tradition of attributing the results of psychological research to individual subjects undergoes a progressive decline. Toward the end of this period most psychological investigations no longer report their results in this way but base their knowledge claims on their use of groups of subjects. In general, the old Wundtian style continues to characterize the areas of psychological research in which it was first introduced, notably the areas of sensation and perception. But in the many new areas of psychological research, both basic and "applied," the new way of establishing psychological knowledge quickly becomes the preferred way.

But this is only a first step in summarizing the historical trend because the categories used here are global in character. Both the category of "individual data" and the category of "group data" are very broad, and cover a number of more specific subcategories. For instance, the "individual data" category could include studies that were truly experimental and presented quantitative data as well as studies that lacked one or both of these features. Actually, this is not a serious problem for this section of the research literature at this particular period because most individual data studies are in fact experimental and quantitative in character. The

Table 5.4. *Percentage of empirical studies reporting individual or group data: Two journals of applied research*

Type of data	*Journal of Applied Psychology*[a]	*Journal of Educational Psychology*
1914–1916		
Individual data only	15	11
Individual and group data	7	14
Group data only	77	75
1924–1926		
Individual data only	11	5
Individual and group data	6	4
Group data only	83	91
1934–1936		
Individual data only	5	3
Individual and group data	2	3
Group data only	93	94

[a]The *Journal of Applied Psychology* only began to appear in 1917; the 1914–1916 analysis actually covers volumes published between 1917 and 1919.

heterogeneity of the "group data" category is much more serious. As we have seen, this category might include such diverse groups as those constituted by criteria like age and sex on the one hand and criteria like mental-test performance on the other. Because other forms of group data have not yet been discussed, we need to extend our analysis to a categorization of the different kinds of groups to which psychological data were attributed. This will provide information on more specific trends in the development of investigative practice. Such information is required for a valid assessment of the historical significance of the global trend.

Further analysis along these lines is also needed for the development of a more differentiated perspective on the relationship between the applied and the basic research literature. In general terms it is certainly true that the "applied" literature anticipates developments in the basic research literature as far as this aspect of investigative practice is concerned. In this respect the terms "applied" and "basic" are particularly misleading if they are taken to imply some historical priority of "basic" research whose results are subsequently "applied" in the field. The truth is that psychological research intended for direct practical application developed its own investigative style, which was very different from what had been the traditional style of basic laboratory research.[31] If anything, the lines of influence seem to have operated in the opposite direction. Historically, it was the investigative practice reported in journals of basic research that came to resemble "applied" research in certain respects, rather than the other way around. We will return to this issue in chapters 7 and 8.

Types of collective subjects

The collective subjects that increasingly became the carriers of psychological knowledge claims were obviously not all of the same kind. Fundamentally different defining characteristics can be used to constitute a collectivity. In this section we will consider these defining characteristics more systematically and trace the use of the different kinds of collective subjects that they constitute in the psychological research literature of the early twentieth century.

In distinguishing among the various kinds of collective subjects that have featured in psychological knowledge claims, the most obvious division to make is that between those collectivities that are the result of the investigator's intervention and those that exist irrespective of this intervention. Laboratory science generally constructs its own artificial objects which do not exist in nature, like pure chemical elements and compounds, whereas field science works with naturally occurring objects. An analogous distinction can be made in connection with psychological collectivities. Some are the artificial products of research practice, whereas others exist anyway. Examples of the latter would be age and sex groups, while the former would include the familiar experimental treatment groups. However, it must be emphasized at once that the "natural" groups of psychological research, in the period being examined here, were not usually natural in quite the same sense as the objects of nonhuman field sciences. The "natural" groups of psychological research generally represented social categories that were of great importance in everyday life. They were a salient part of social life outside the laboratory. As a result, psychological research on groups representing such categories tended to have rather direct social implications.

The natural groups of psychological research were taken to represent categories that were important in the organization of social life outside the laboratory. For example, a study that attributed certain psychological characteristics to a group of children defined by their age would have direct implications within a social order marked by an age-graded system of compulsory formal education. Groups representing male and female categories would have an analogous social significance. Psychologists did not create these categories; they simply took them over from the social order of which they were a part. Their research was simply designed to add further information or to specify more accurately the attributes of categories that were accepted without question. The collective subjects of this kind of psychological research were not defined by psychological considerations but by cultural givens.

A very different situation existed where psychologists constructed the collective subjects of their research in terms of their own criteria. In this case the population about whom the research process provided specific information was not related to any particular social category but was simply a product of the researchers' intervention. The earliest examples of this occur when the experimental responses of a number of subjects are averaged to yield a group mean. Innocent though such a procedure seems at first sight, it implies a fundamental conceptual change. Such a group mean is obviously not the attribute of any of the actual persons who contributed to its constitution but the attribute of a collectivity. But what kind of collectivity is this? It is the group of individuals who happened to participate as sources of data in a particular psychological investigation. Their common activity in the experimental situation defines them as a group. There never was such a group before the investigation took place. They do not represent any preexisting social category. If this group represents anything at all, other than itself, it is a hypothetical population of individuals who could all be subjected to the same investigative procedures. In other words, the collectivity of which the group mean is the attribute is one that is defined by laboratory practice rather than by social practice outside the psychological laboratory.[32]

It is difficult to overemphasize the potential importance of this shift for psychology as a discipline. It points the way to a science that supplies its own categories for classifying people and is not dependent on the unreflected categories of everyday life. This has strong implications both for the internal development of the discipline and for its social impact. As regards the latter, there is now the possibility that psychologically constituted collectivities will compete with and even replace some of the more traditional categories of social grouping. Categorizing children or recruits by intelligence quotient was an early example of such a process.[33] In terms of the internal development of the discipline, the artificial constitution of collectivities by the research process itself provided the basis for a science of psychological abstractions that need never be considered in the context of any actual individual personality or social group.

There are three major ways in which the psychologist's intervention produced new groupings of individuals that did not correspond to groupings on the basis of traditional social criteria. The first way occurs when a number of individuals are subjected to analogous experimental treatment and their responses are pooled. A learning curve based on the average responses of a number of subjects would be an example of this process. The relationship between some response measure and the number of trials, which such a curve illustrates, is attributed to a group of subjects and is not expected to be empirically demonstrable in every (or perhaps any) individual member of the group. Such a group is purely an artifact of experimental practice. What defines membership in such a group is common exposure to a particular pattern of experimental intervention. Presenting the average results for the group instead of the results for each individual suggests that the attributed function is like a general natural scientific law that holds across individual subjects.[34]

A second kind of artificial group is created when the individual subjects in an experiment are deliberately subdivided on the basis of their differential exposure to different experimental conditions. Whereas in the first case the responses of all the subjects who participated in the experiment are combined after the event, we now get a preplanned division of the subjects into two or more groups, each exposed to a different pattern of experimental treatment. What is now of interest is not so much the actual pattern of group response as the differences between these patterns in different groups. Such differences can then be causally attributed to differences in the experimental treatment of the groups. In other words, we are dealing with the kinds of groups that are the product of what has come to be known as control-group design. Here the groups to which experimental results are attributed are defined by the differential experimental treatment to which their respective members are exposed. We may therefore call them *treatment groups* to distinguish them from the *general experimental groups* discussed in the previous paragraph. Both kinds of groups are artifacts of experimental practice, but they must be distinguished from each other because they have a different history, as we shall see, and because they imply a different approach to experimentation.

A third kind of artificial group produced by psychological investigation is the *psychometric group*. In this case the defining characteristic of group membership is based on performance on some psychological measuring or assessment instrument. In the historical period examined here this is usually a psychological test, most often an intelligence test, but it could equally well be an attitude scale or personality questionnaire. Groups defined by certain levels of performance on such instruments are obviously the product of the psychologist's intervention and are not "natural" groups based on some preexisting social classification. Psychometric groups differ from the other artificial collectivities we have discussed in the different interpretation that is given to the psychologist's intervention. In the case of general experimental and treatment groups, the intervention that defined the group was thought of as a modifying intervention, but in the case of psychometric groups the defining intervention was supposed to be one that elicited a stable characteristic of group members.

Given that in any investigative situation the responses of psychological "subjects" will depend both on what they bring to the situation and on what the situation brings to them, all three kinds of artificial groups are artificial in two senses. The first sense is the one we have already discussed; it expresses the fact that such groups are created in the process of investigation and do not exist in the everyday world outside this process. But in creating such groups, psychologists also made them artificial in a second sense. If we regard the noncorrespondence of these groups to preexisting social categories as a kind of sociological artificiality, then we also should take notice of a fundamental artificiality on the psychological level. The existence of experimental and treatment groups was defined solely in terms of certain modifying factors in the investigative situation, and more or less stable characteristics of individuals were relegated to the category of "error." Psychometric groups, however, were defined solely in terms of supposedly stable characteristics of individuals, and situational effects were either ignored or relegated to artifactual status. Thus, in every case the defining characteristic that constituted the investigated group was based on a psychological abstraction or idealization. In the one case the idealization was that of a collective organism that exhibited only modifiability; in the other case it was that of a collective organism that exhibited only stable traits.[35]

This distinction was important for the kinds of knowledge claims that psychologists were interested in making. In order to make universalistic knowledge claims psychologists took to presenting their data as the attributes of collective rather than individual subjects. Very frequently, these collectivities were constructed by psychologists for this specific purpose, and in constructing them they postulated the existence of a collective organism that already exhibited the assumed general characteristics on which their knowledge claims depended. Thus, to demonstrate the effects of supposedly stable characteristics, they constructed experimental groups defined by such assumed characteristics, and to demonstrate the modifying effects of experimental intervention they constructed groups entirely defined by exposure to such intervention. Such a procedure was inevitable if knowledge claims were to be made through the medium of a myriad of separate, relatively small-scale empirical studies. Problems only arose when the hypothetical nature of the collective subjects that featured in psychological investigations was forgotten, and attempts were made to transfer the results of such investigations to the world of real subjects.[36]

6

Identifying the subject in psychological research

Human sources for psychological generalizations

Making psychological knowledge claims, we saw in the last chapter, involves an ordered set of attributions attached to a particular kind of subject. The significance of these attributions depends in a crucial way on the nature of the subject whose attributes they are claimed to be. If the subject is an individual consciousness, we get a very different kind of psychology than if the subject is a population of organisms. The gradual predominance of the latter type of subject was seen to depend on the interest of psychologists in a type of knowledge that was applicable to populations outside the laboratory but that was as abstract as the generalizations of the physical sciences. This meant working with artificially defined collective subjects in the research situation and then imposing these definitions on people outside the laboratory.

In other words, psychological research on populations had a tendency to replace the social categories that defined populations in real life with populations defined in terms of nonsocial categories. American psychology aimed to be a socially relevant science, but not a social science. Its approach was to be that of a natural science, although its ultimate field of application was to be found among members of real societies.[1] Thus, research on socially defined populations came to be regarded as "applied" research, whereas "basic" research worked with abstract populations. In the extreme case the populations need not even be human. In fact, white rats fulfilled the role of a population of completely abstract organisms more adequately than any human population could hope to do.[2]

But where human populations were retained, the making of appropriately general knowledge claims faced a problem: Each empirical investi-